CEE 465 & CEWA 565

Data Analysis in Water Sciences

Lecture 15: Intro to SVD / EOF / PC

Nov. 24, 2020

- If you need a different presentation date, find someone to trade with!
- Students taking the class asynchronously have the option to pre-record their presentations

CEWA 565 Project Presentation Dates

Presentation Date	Project Group
Thursday, Dec 3rd	Caden Chamberlain, Anthony Stewart
	Shannon Cavanaugh, Jessica Steigerwald
	Yasmeen Mansour, Fanny Okaikue-Woodi
	Leigh Hamlet
	Hannah Besso, Eli McMeen
Tuesday, Dec 8th	Erik Dahl, Emily Iseley
	Craig Dittmann, Marek Kossik, Christian Vanderhoeven
	Ben Roberts, Zheng Luo
	Bareera Nadeem, Devin Robichaux*
	Sitao Liu, Zhuoran Li, Hongyi Yu*
Thursday, Dec 10th	Allan Cortina
	Kory Swabb
	Tara Okahara, Kori Chun
	Chang He, Will Wu
	*prerecorded presentations

See grading rubric:
https://mountain-hydrology-research-group.github.io/data-analysis/overview/b-project.html

Announcements

CEWA 565 Project Tips:

- Why should your audience care?
- What about your physical system of interest led you to choose the statistical test(s) you did?

Announcements

CEWA 565 Project Tips:

Your research questions have two parts

Real world, physical part: Based on your understanding of the world, your system, your data, what do you want to know? (This should help us better understand the world, or make better engineering decisions.)

Statistical part: This is where you choose a specific statistical test or analysis (or several) that you perform to try to provide objective insight into the first part.

Announcements

Today:

- Singular Value Decomposition (Lab 8-1)

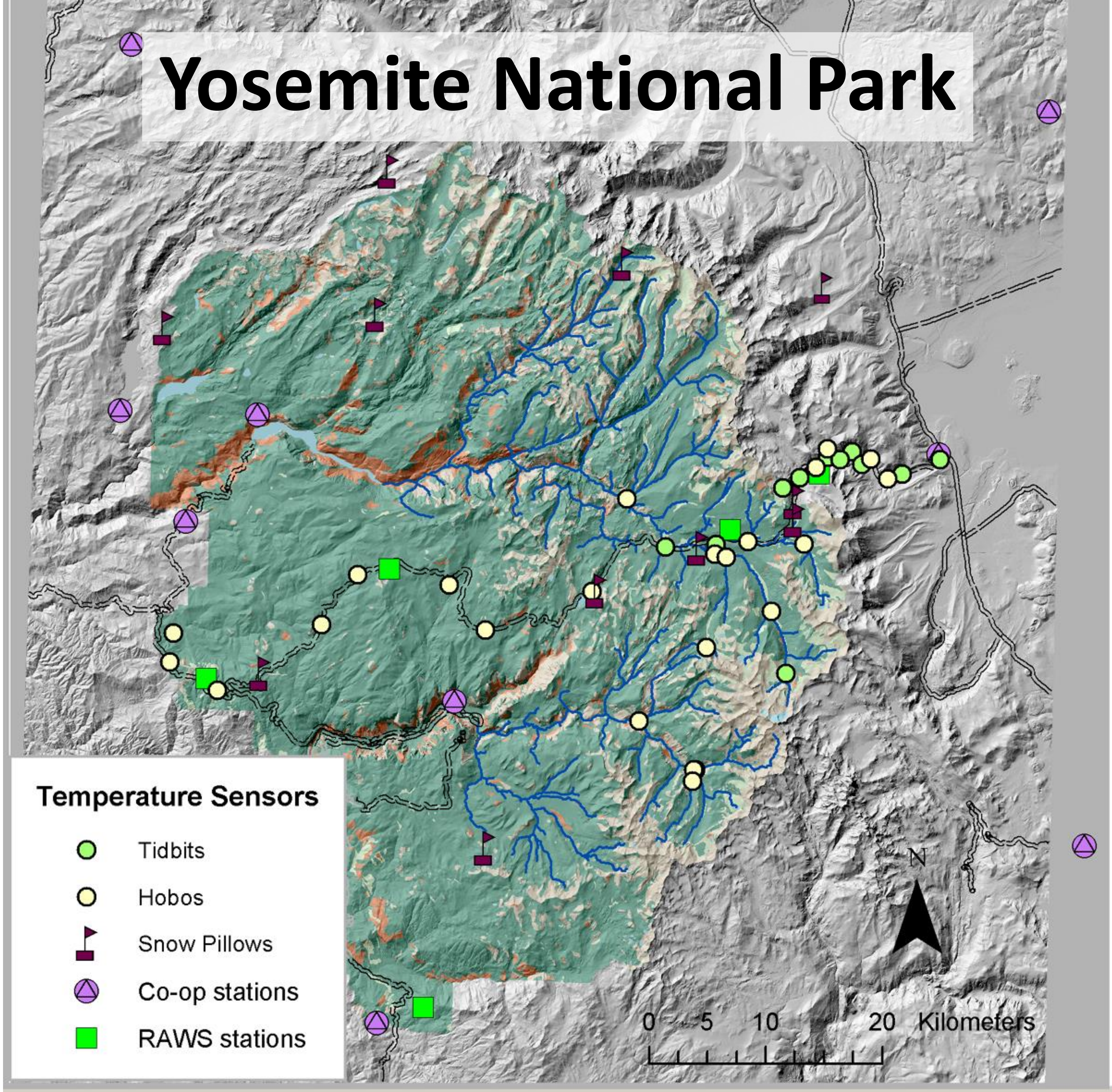
Homework 7: Due <u>Tuesday</u> Dec. 1st, 10:30 AM (Pacific Time)

- Problem 1: MCMC (from module 7)
- Problem 2: SVD (from module 8)
- Problem 3: Statistics Synthesis (CEE 465)

Intro to Singular Value Decomposition (SVD), Empirical Orthogonal Functions (EOF), Principal Components (PC)

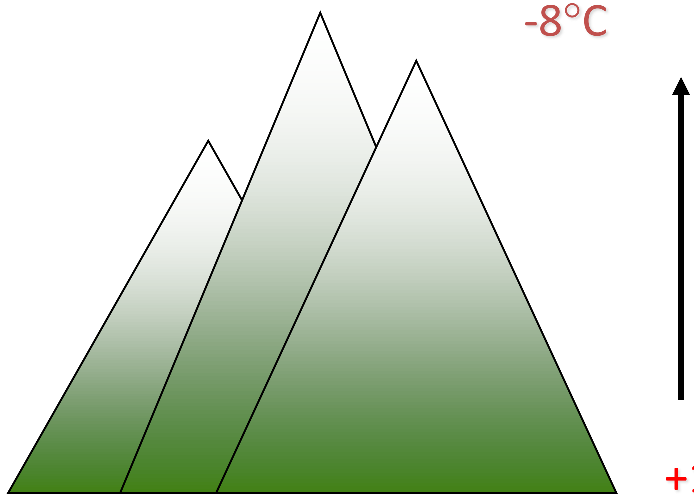
(applications oriented, less theory)

Examples derived from: Lundquist and Cayan, JGR (2007) and Lundquist et al., JGR (2008)


Yosemite National Park
Temperature Sensors
Tidbits
Hobos
Snow Pillows
Co-op stations
RAWS stations
N
0 5 10 20 Kilometers


YOSEMITE
N.P.
CALIFORNIA

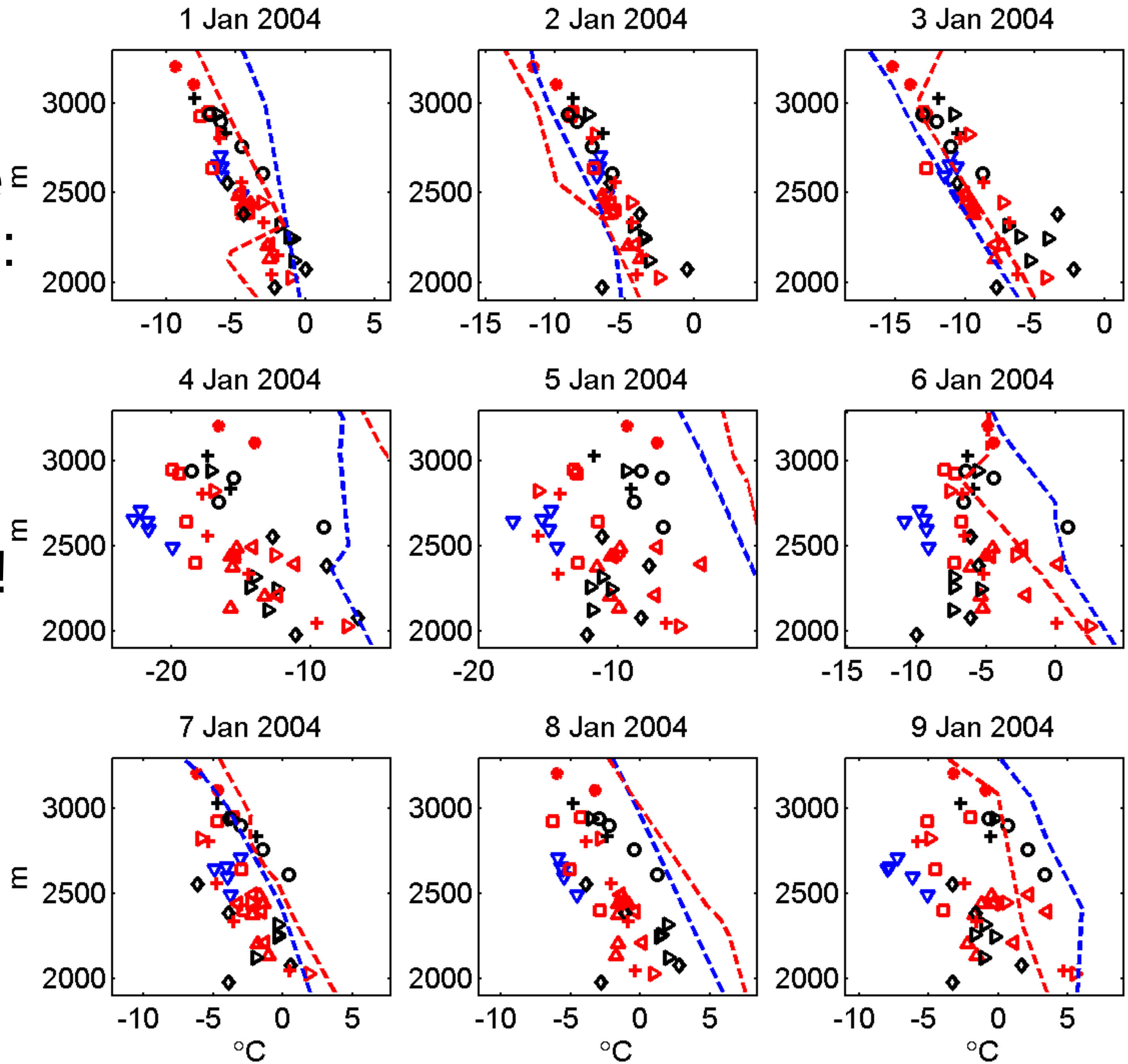
In Standard Atmosphere, Temperature Decreases with Altitude



Average decrease = 6.5 °C / km

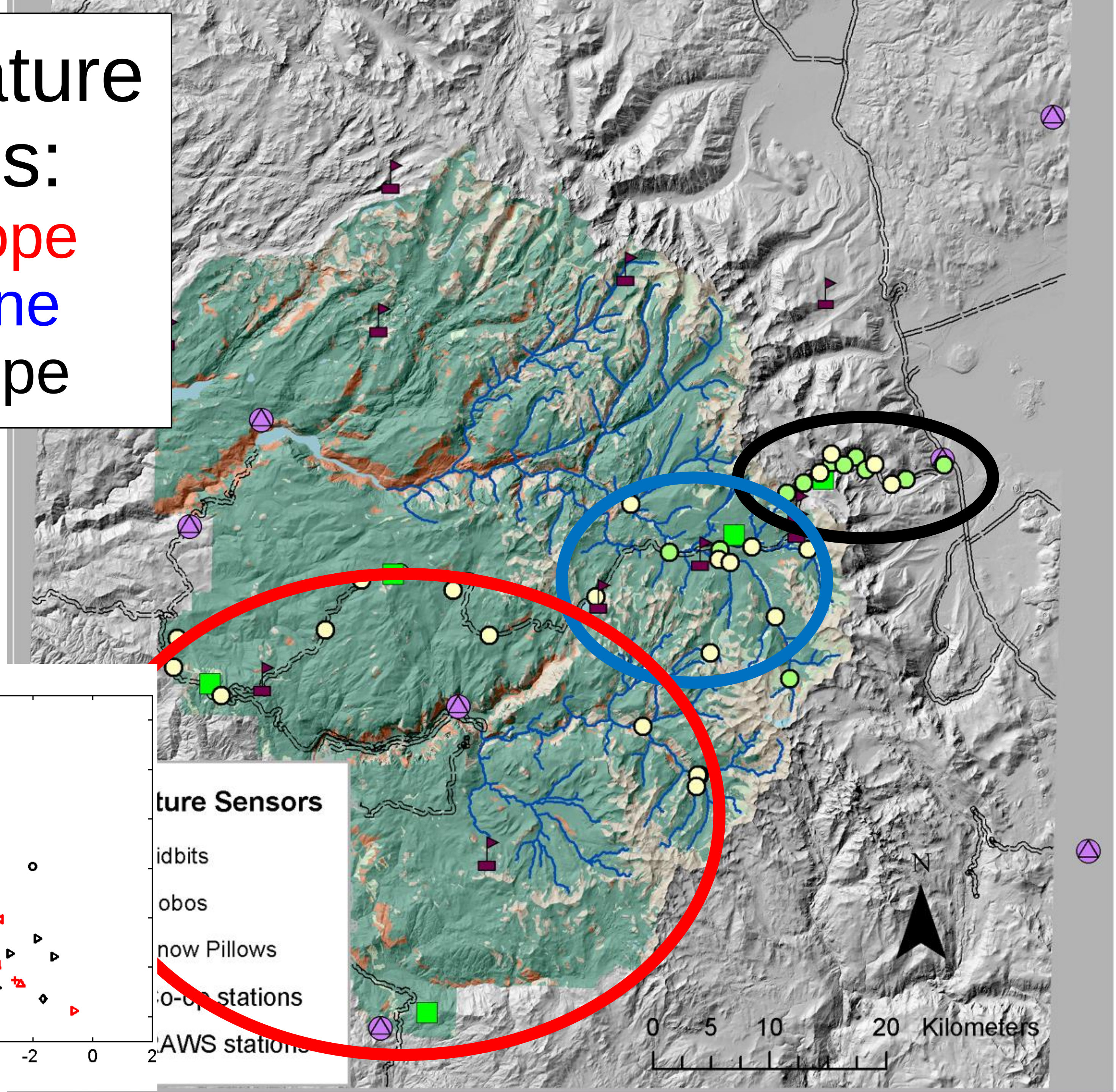
(linear model, this is used for most snowmelt models)

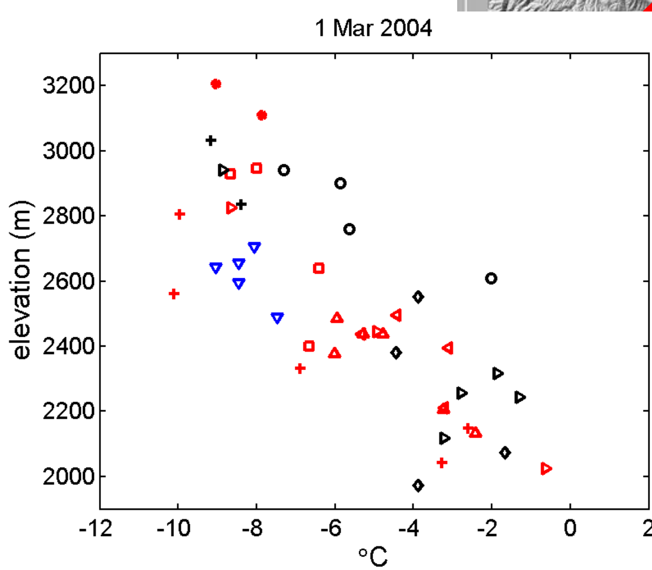
Yosemite temperature vs. elevation: not always linear or related to free atmosphere!



Symbols are different ground temperature sensors. Dashed lines are from balloon soundings.

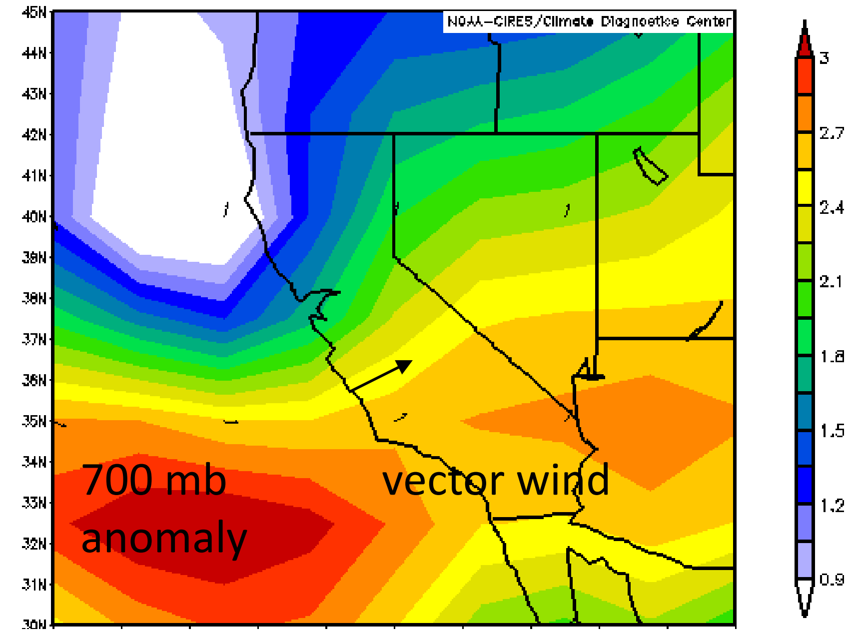
Temperature Groups:
West-slope
Tuolumne
East-slope

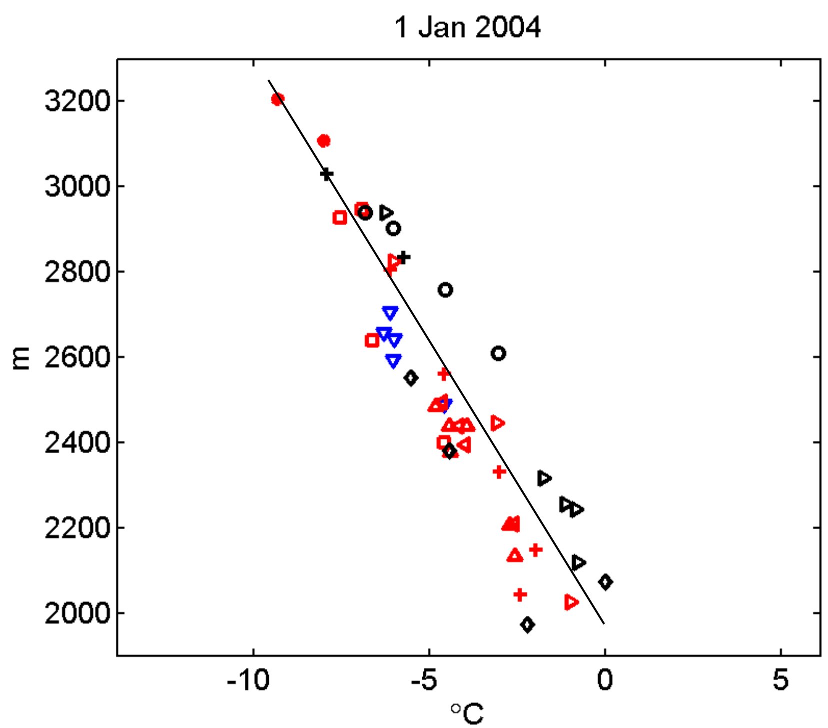


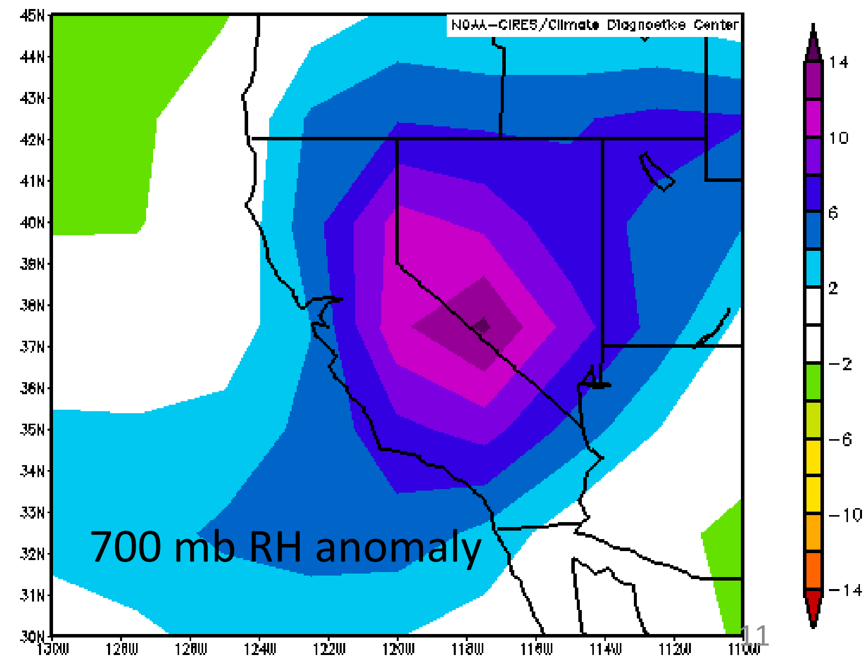


"Perfect" lapse rate

- Strong winds, directly perpendicular to range
- Usually cloudy, stormy trough
- Free atmosphere dominates, can neglect local topography

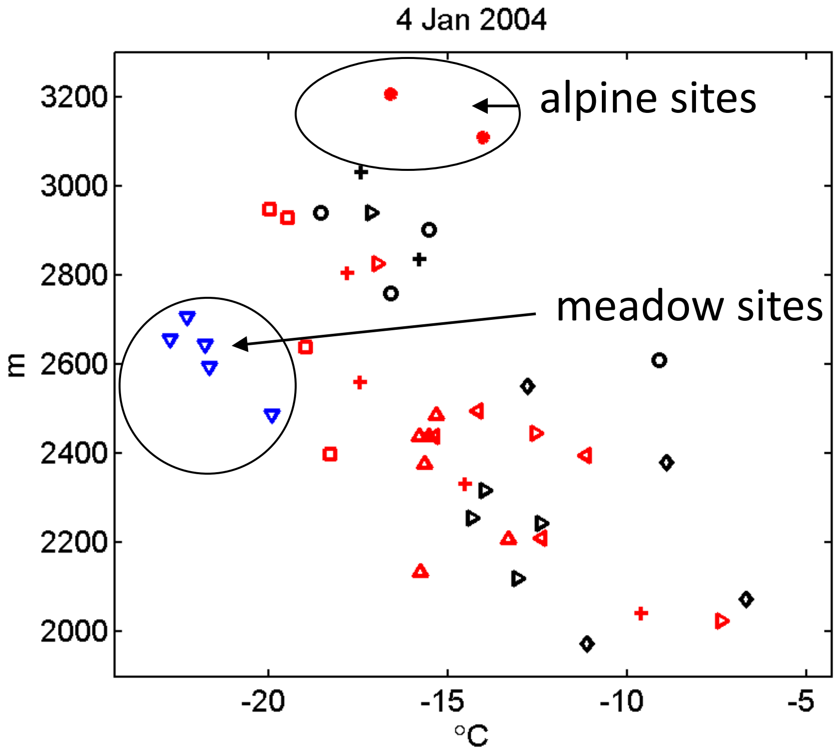


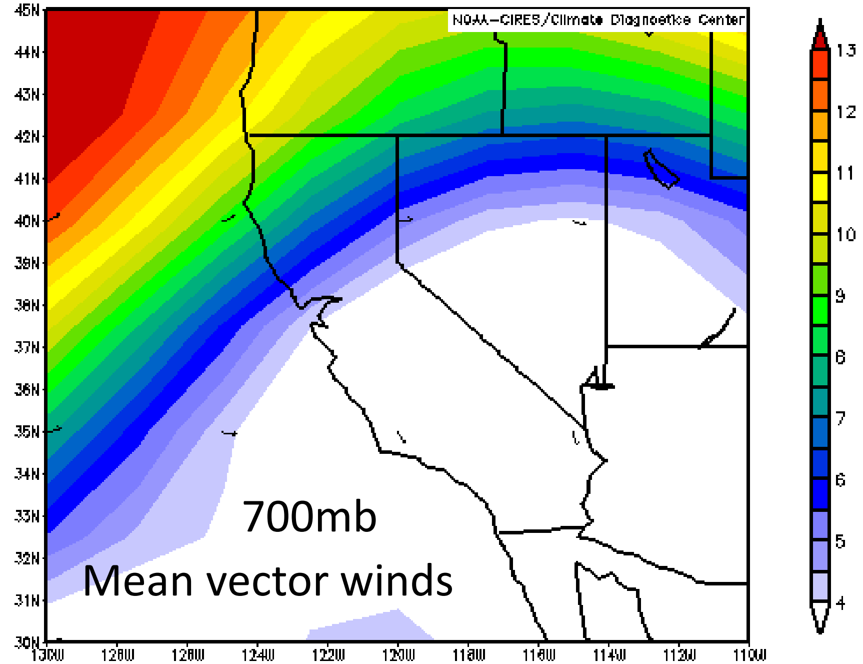


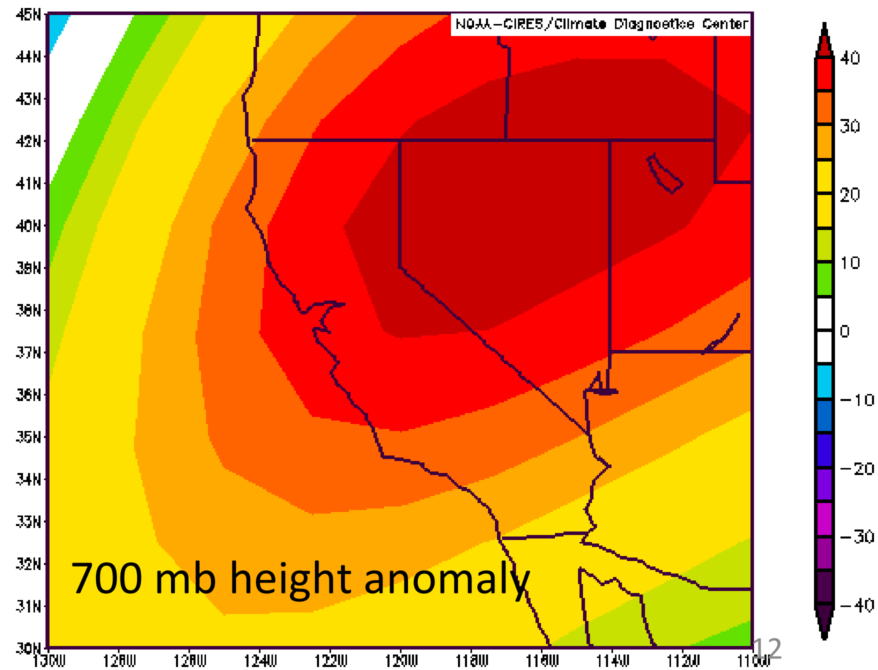


Inversion

- High pressure/ridge
- Sinking air, weak winds
- Cannot define linear lapse rate
- Local topography most important

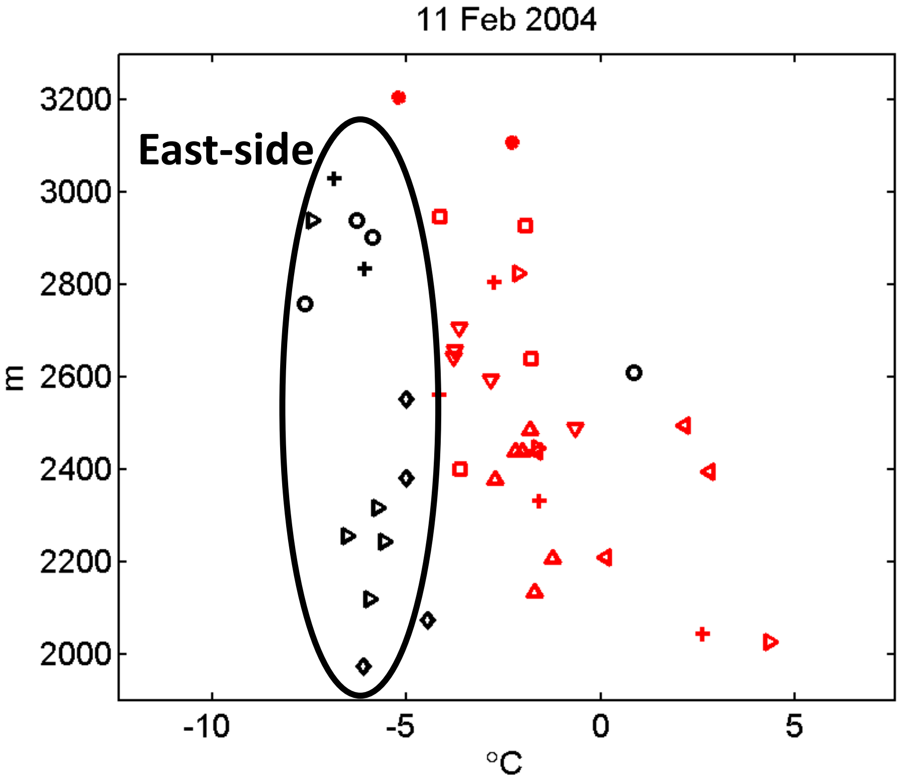


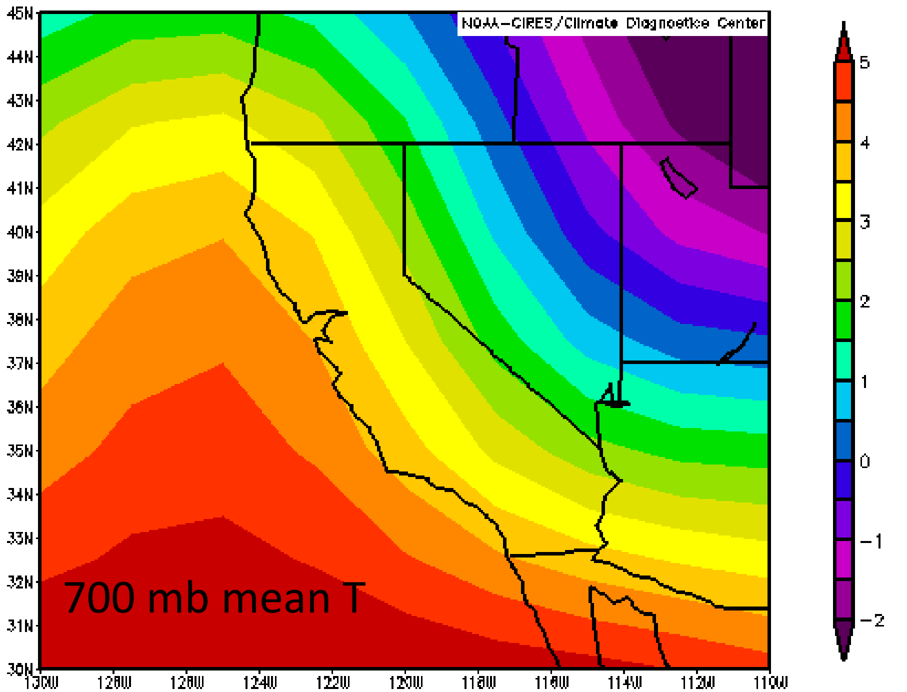


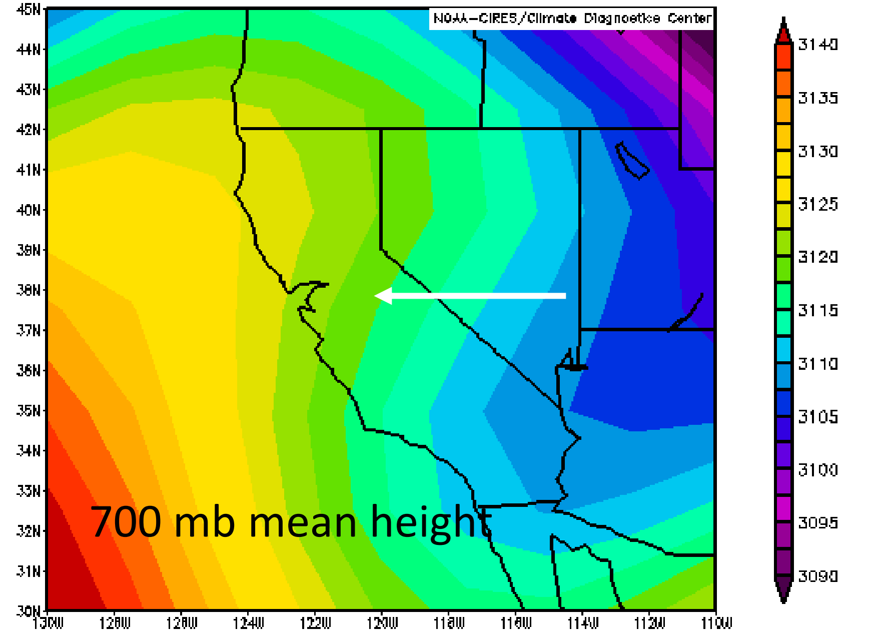


T_east < T_west

- Eastern slopes much colder than western

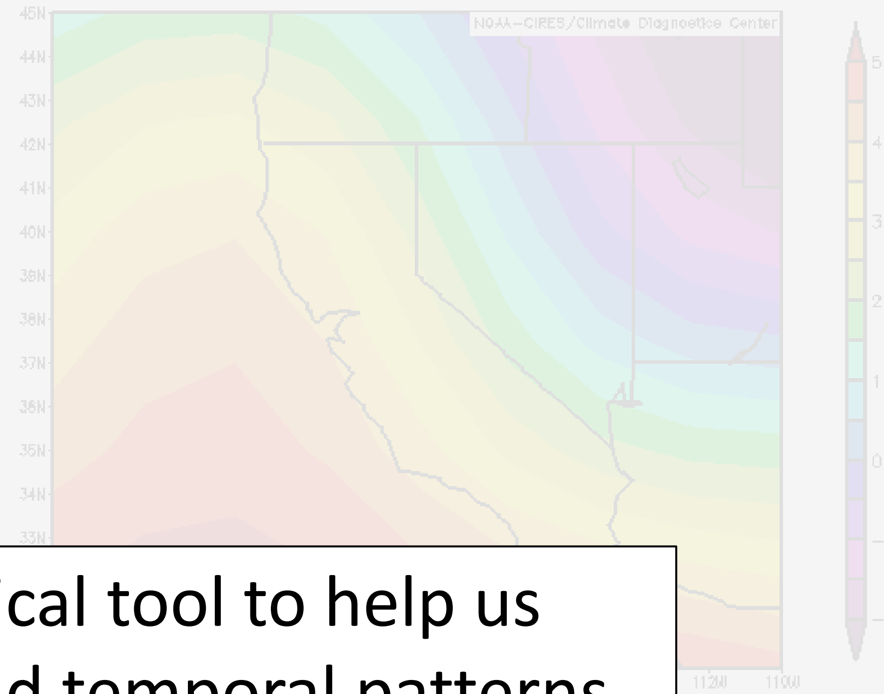


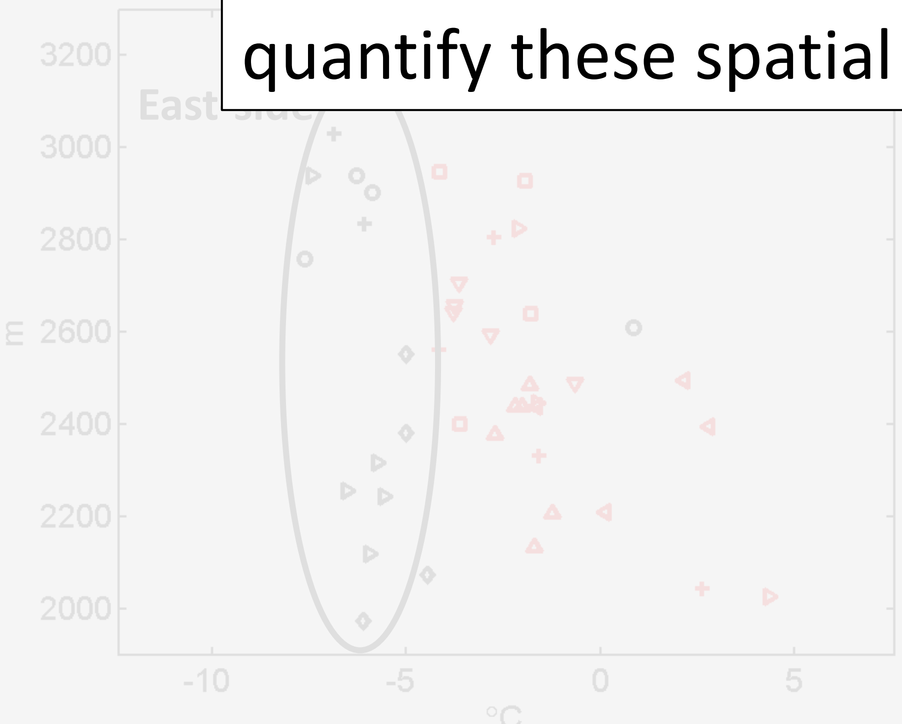




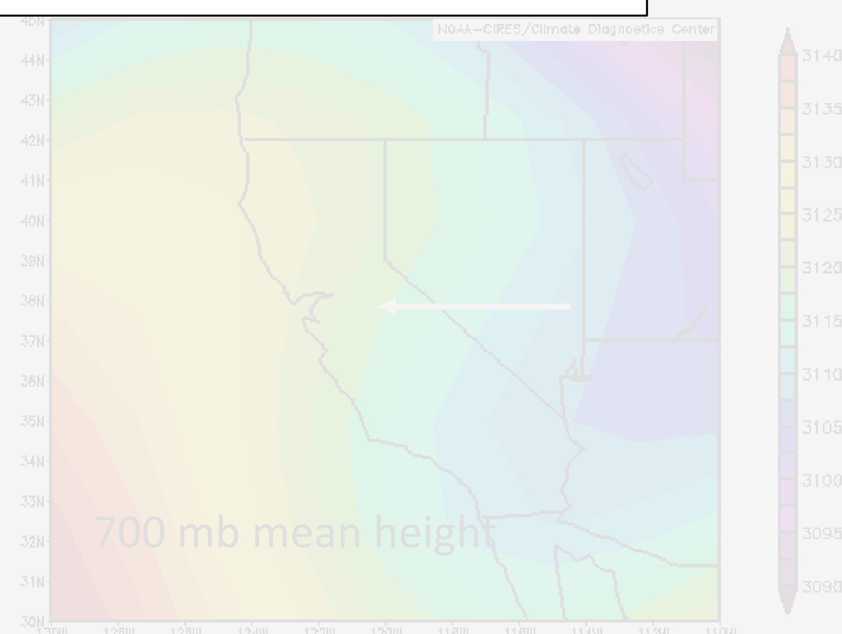
T_east < T_west

- Eastern slopes much colder than western





We need a better statistical tool to help us quantify these spatial and temporal patterns.



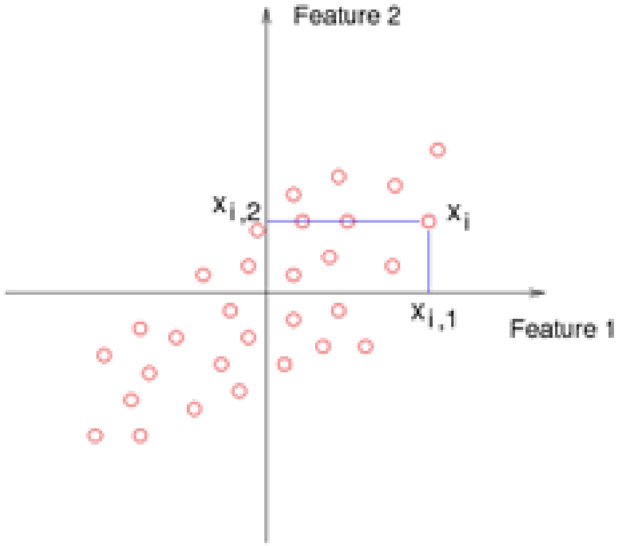
Analysis Method:
Empirical Orthogonal Functions (EOF's)

-Each EOF consists of a spatial weight (EOF) and a temporal weight, or principal component (PC)

-Spatial weights represent primary modes of variability over the study area

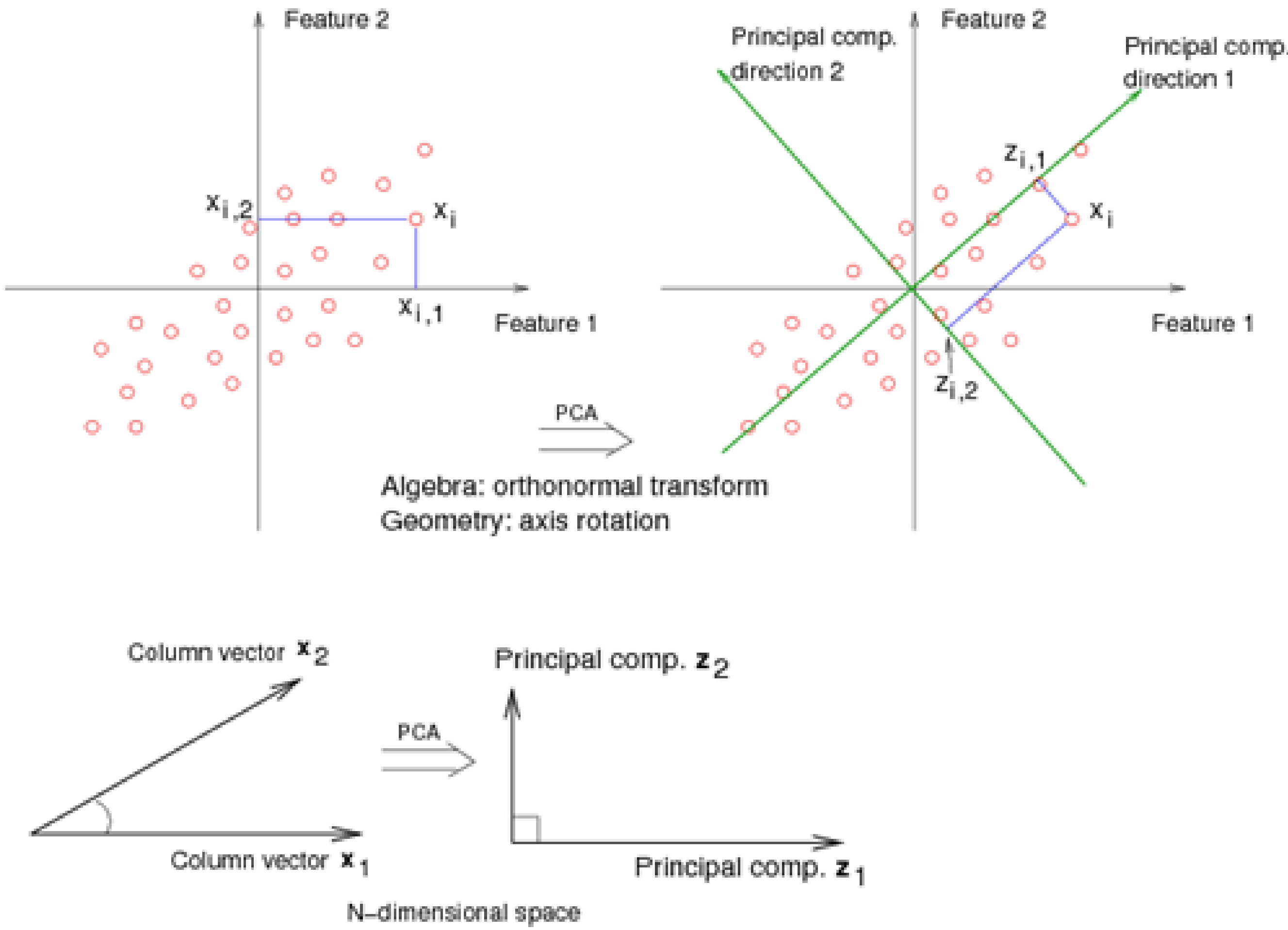
-Temporal weights explain how those modes change in time

Graphical Representation:
What are the primary directions of variability? Rotate the axis to match those directions.



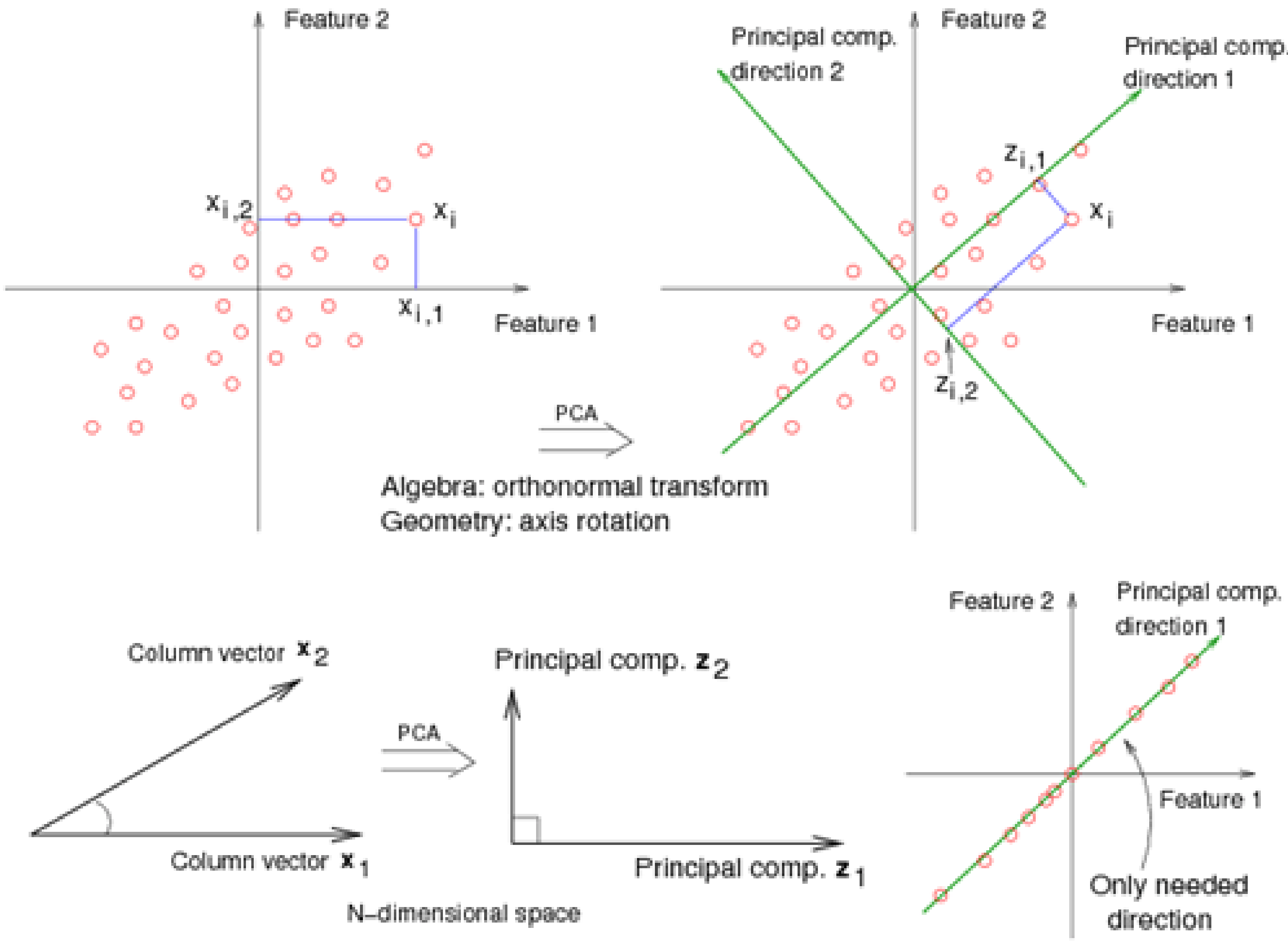
Graphical Representation:

What are the primary directions of variability? Rotate the axis to match those directions.



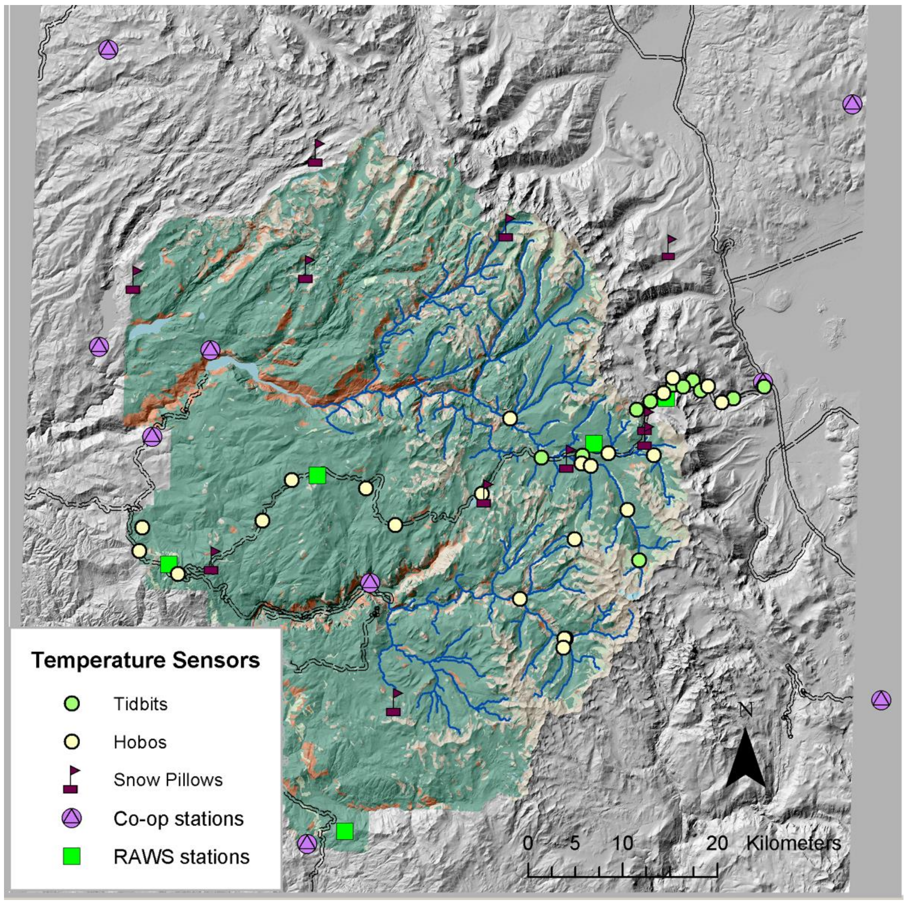
Graphical Representation:

What are the primary directions of variability? Rotate the axis to match those directions.



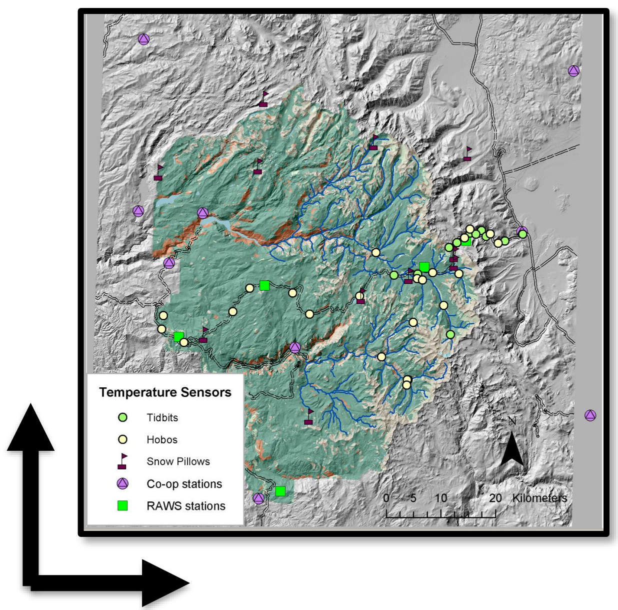
Empirical Orthogonal Functions (EOF's)

With mountain air temperature

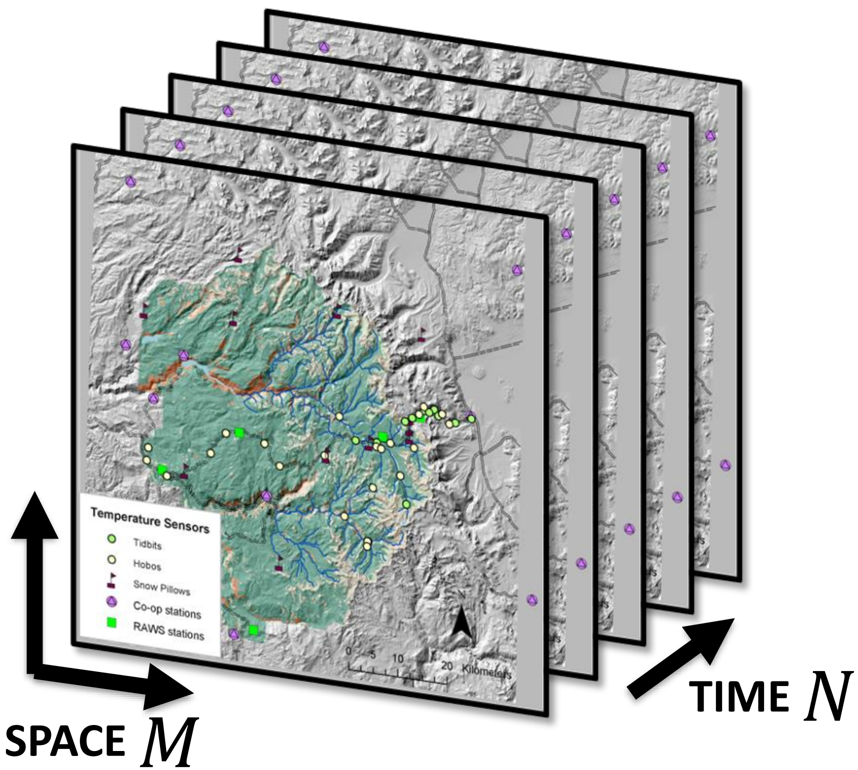


Steps:

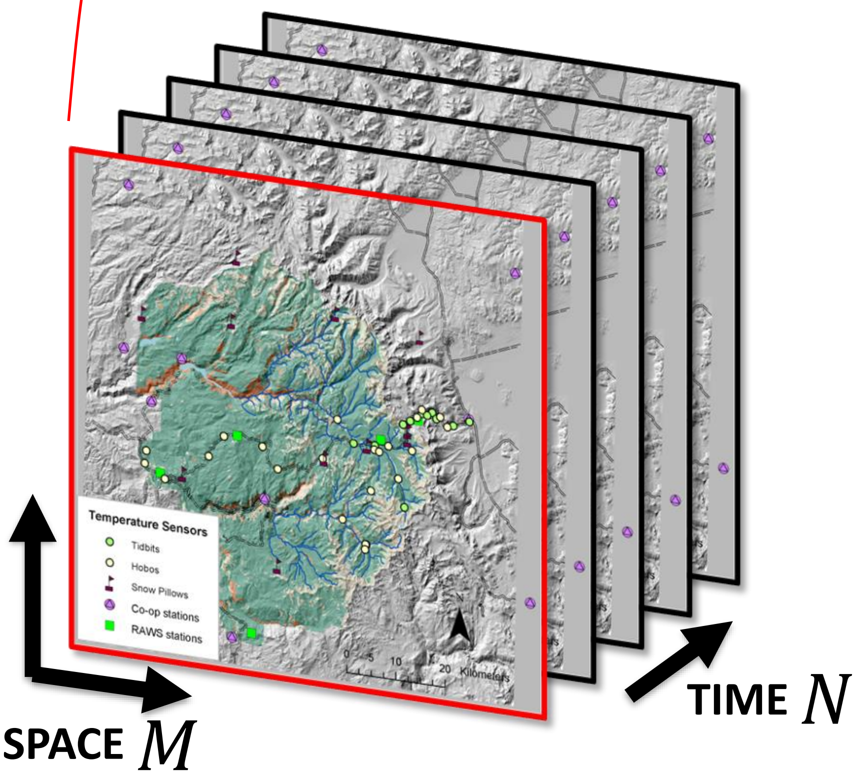
1. Arrange data into matrix
2. Remove mean from data
3. Singular Value Decomposition
4. Interpretation


Temperature Sensors
Tidbits
Hobos
Snow Pillows
Co-op stations
RAWS stations
0 5 10 20 Kilometers
N

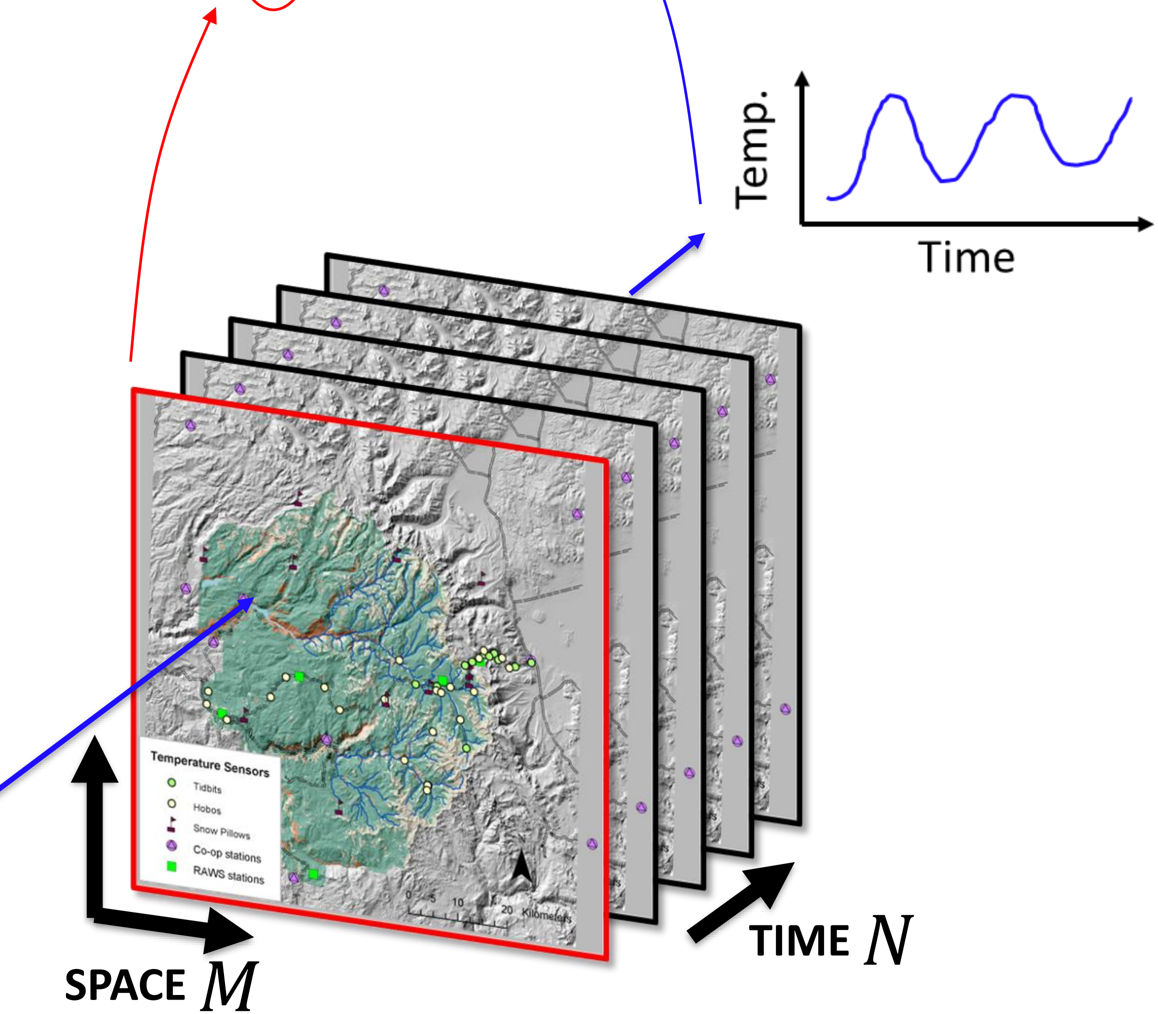
SPACE M


Temperature Sensors
Tidbits
Hobos
Snow Pillows
Co-op stations
RAWS stations
SPACE M
TIME N

$$\boldsymbol{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,N} \\ \dots & \dots & \dots \\ x_{M,1} & \dots & x_{M,N} \end{bmatrix}$$

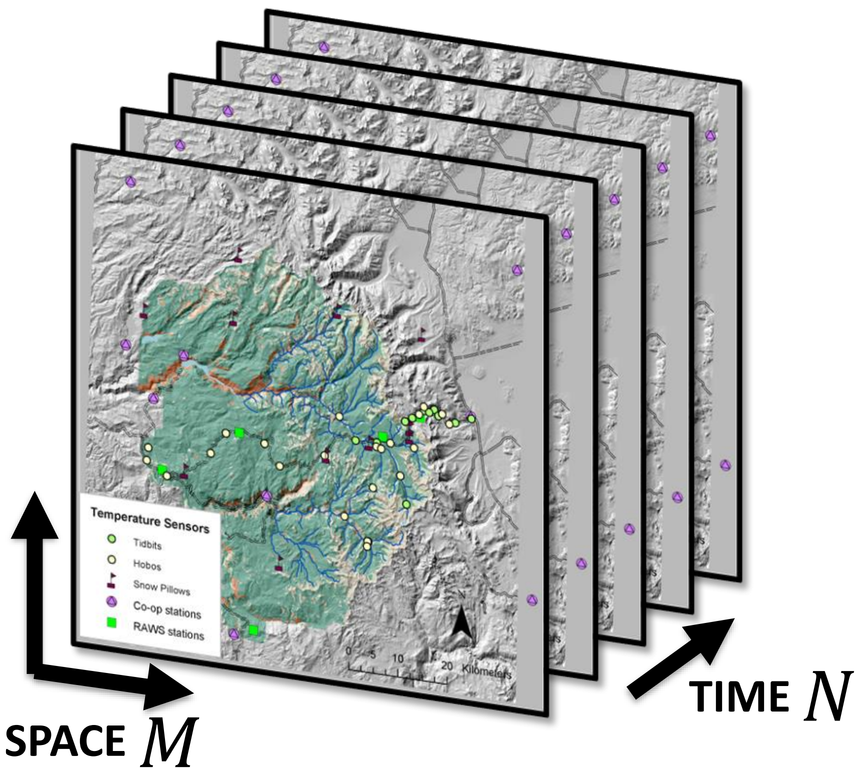


$$\boldsymbol{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,N} \\ \dots & \dots & \dots \\ x_{M,1} & \dots & x_{M,N} \end{bmatrix}$$



$$\boldsymbol{X} = \begin{array}{cc} N & \\ \begin{bmatrix} x_{1,1} & \dots & x_{1,N} \\ \dots & \dots & \dots \\ x_{M,1} & \dots & x_{M,N} \end{bmatrix} & M \end{array}$$

Where $\boldsymbol{X} = \boldsymbol{X} - \bar{\boldsymbol{X}}$



$$\begin{array}{cc} & N \\ \boldsymbol{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,N} \\ \dots & \dots & \dots \\ x_{M,1} & \dots & x_{M,N} \end{bmatrix} & M \end{array}$$

Covariance of spatial locations with one another. "Dispersion matrix" with N, time, as the sampling dimension.

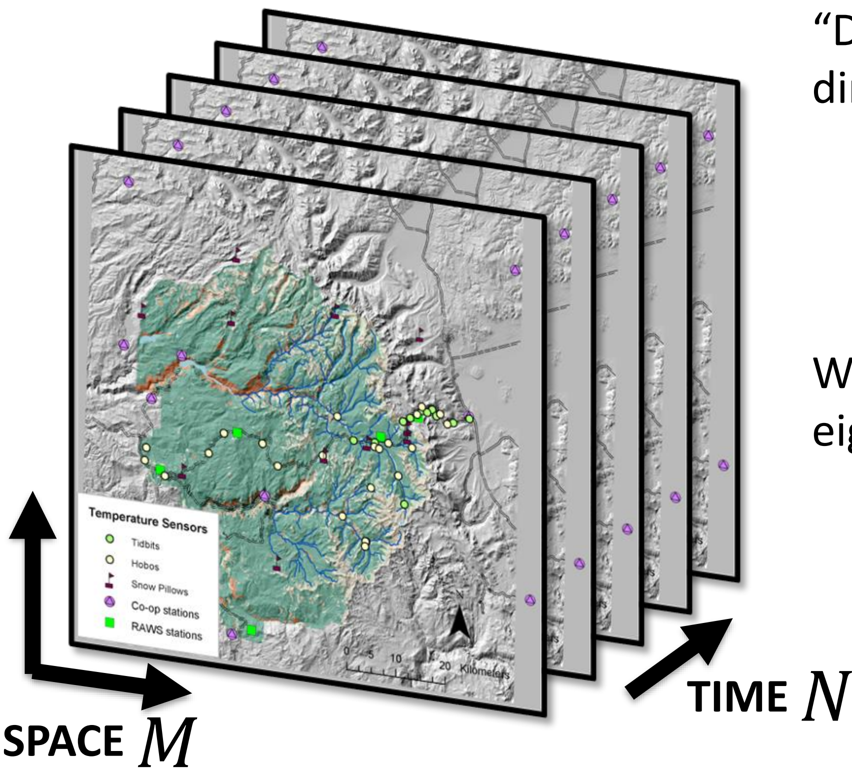
$$\boldsymbol{C} = \boldsymbol{X}\boldsymbol{X}^T = \begin{array}{c} N \\ \left[\quad\right] M \end{array} \begin{array}{c} M \\ \left[\quad\right] N \end{array} = \begin{array}{c} M \\ \left[\quad\right] M \end{array}$$

Covariance of different times with one another. "Dispersion matrix" with M, space, as the sampling dimension.

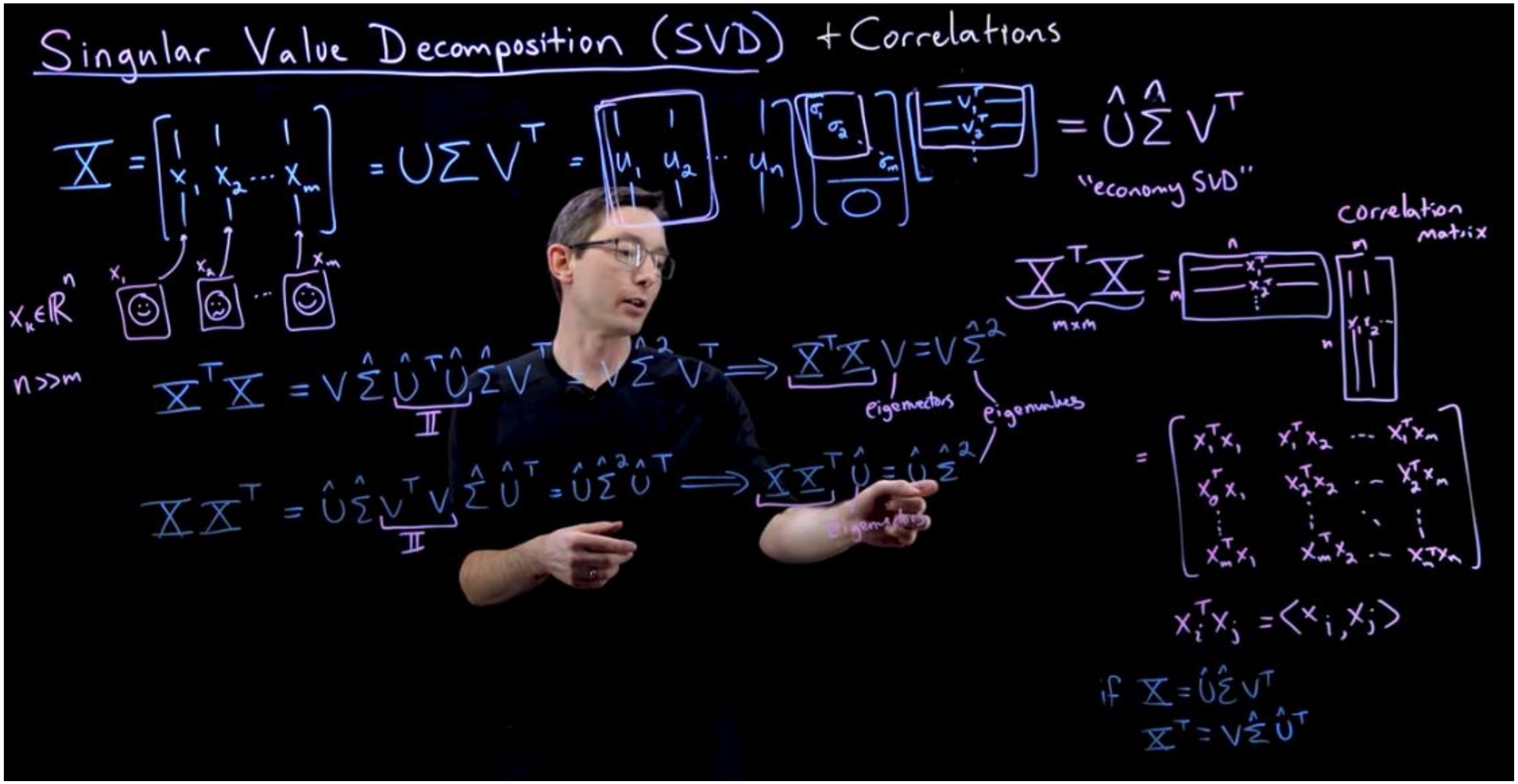
$$\boldsymbol{C} = \boldsymbol{X}^T\boldsymbol{X} = \begin{array}{c} M \\ \left[\quad\right] N \end{array} \begin{array}{c} N \\ \left[\quad\right] M \end{array} = \begin{array}{c} N \\ \left[\quad\right] N \end{array}$$

We want to solve for the eigenvectors ($\boldsymbol{v}$) and eigenvalues (λ) of our covariance matrices.

$$\boldsymbol{C}\boldsymbol{v} = \lambda\boldsymbol{v}$$



Steve Brunton, Singular Value Decomposition



https://www.youtube.com/playlist?list=PLMrJAkhIeNNSVjnsviglFoY2nXildDCcv

Singular Value Decomposition (SVD):

$$\boldsymbol{X} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,N} \\ \cdots & \cdots & \cdots \\ x_{M,1} & \cdots & x_{M,N} \end{bmatrix} = \boldsymbol{USV}^{T}$$

In python:

```
U, S, V = scipy.linalg.svd(X)
```

Singular Value Decomposition (SVD):

$$\boldsymbol{X} = \begin{bmatrix} x_{1,1} & \dots & x_{1,N} \\ \dots & \dots & \dots \\ x_{M,1} & \dots & x_{M,N} \end{bmatrix} = \boldsymbol{USV}^T = \overset{\boldsymbol{U}}{\begin{bmatrix} \dots & \dots & \dots \\ u_1 & \dots & u_M \\ \dots & \dots & \dots \end{bmatrix}} \overset{\boldsymbol{S}}{\begin{bmatrix} s_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & s_N \end{bmatrix}} \overset{\boldsymbol{V}^T}{\begin{bmatrix} \dots & \dots & \dots \\ v_1 & \dots & v_N \\ \dots & \dots & \dots \end{bmatrix}^T}$$

EOF's

- "Spatial eigenvectors" (MxM)
- Eigenvectors of XX^T
- (left singular values)

In python:

```
U, S, V = scipy.linalg.svd(X)
```

Singular Value Decomposition (SVD):

$$\begin{array}{ccc} & \boldsymbol{U} & \boldsymbol{S} & \boldsymbol{V}^T \end{array}$$

$$\boldsymbol{X} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,N} \\ \cdots & \cdots & \cdots \\ x_{M,1} & \cdots & x_{M,N} \end{bmatrix} = \boldsymbol{USV}^T = \begin{bmatrix} \cdots & \cdots & \cdots \\ u_1 & \cdots & u_M \\ \cdots & \cdots & \cdots \end{bmatrix} \begin{bmatrix} s_1 & 0 & 0 \\ 0 & \cdots & 0 \\ 0 & 0 & s \end{bmatrix} \begin{bmatrix} \cdots & \cdots & \cdots \\ v_1 & \cdots & v_N \\ \cdots & \cdots & \cdots \end{bmatrix}^T$$

EOF's

- "Spatial eigenvectors" (MxM)
- Eigenvectors of XX^T
- (left singular values)

PC's

- Principal Components (NxN)
- Eigenvectors of X^TX
- (right singular values)

In python:

```
U, S, V = scipy.linalg.svd(X)
```

Singular Value Decomposition (SVD):

$$\boldsymbol{X} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,N} \\ \cdots & \cdots & \cdots \\ x_{M,1} & \cdots & x_{M,N} \end{bmatrix} = \boldsymbol{USV}^T = \overset{\boldsymbol{U}}{\begin{bmatrix} \cdots & \cdots & \cdots \\ u_1 & \cdots & u_M \\ \cdots & \cdots & \cdots \end{bmatrix}} \overset{\boldsymbol{S}}{\begin{bmatrix} s_1 & 0 & 0 \\ 0 & \cdots & 0 \\ 0 & 0 & s_N \end{bmatrix}} \overset{\boldsymbol{V}^T}{\begin{bmatrix} \cdots & \cdots & \cdots \\ v_1 & \cdots & v_N \\ \cdots & \cdots & \cdots \end{bmatrix}^T}$$

EOF's

- "Spatial eigenvectors" (MxM)
- Eigenvectors of XX^T
- (left singular values)

PC's

- Principal Components (NxN)
- Eigenvectors of X^TX
- (right singular values)

Singular values

- Eigenvalues (MxN)
- How much variance is explained by each eigenvector

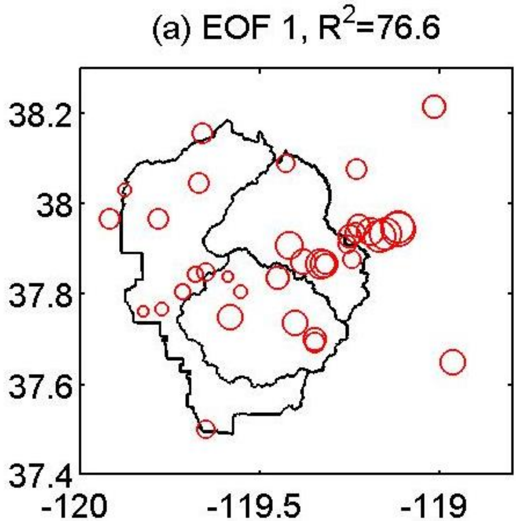
In python:

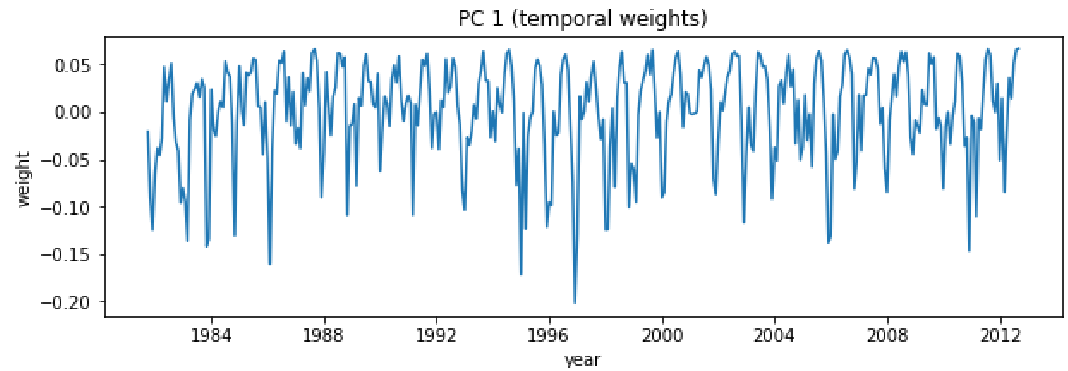
```
U, S, V = scipy.linalg.svd(X, full_matrices=False)
```

Spatial Weights (EOFs)

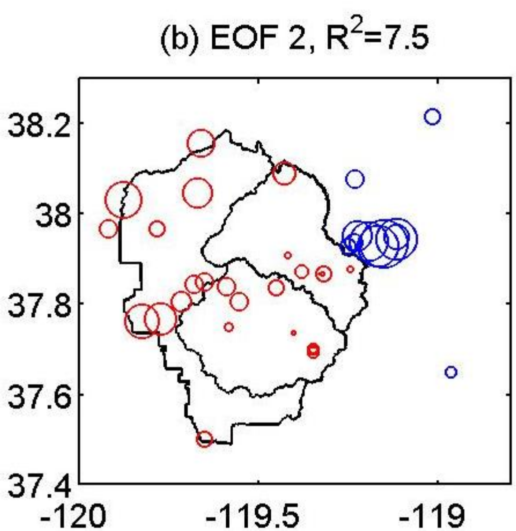
Temporal Weights (PCs)

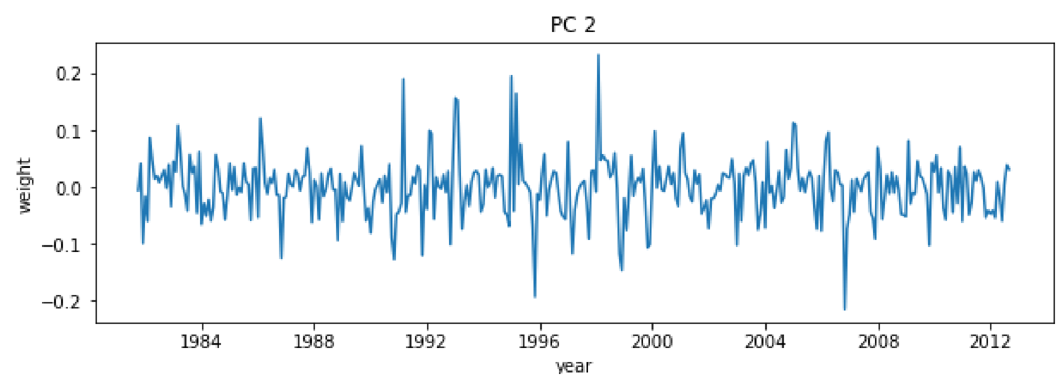
Region-wide





East-vs-West

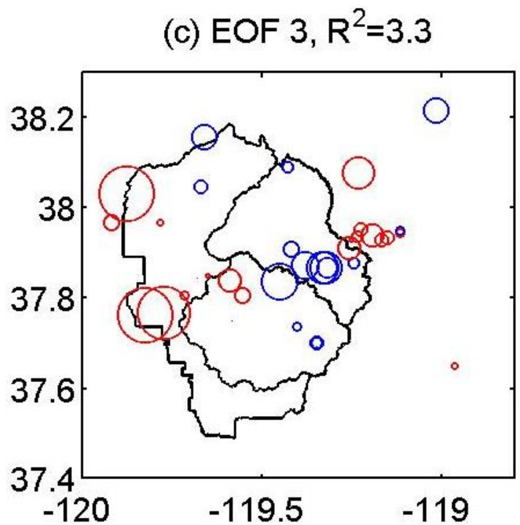


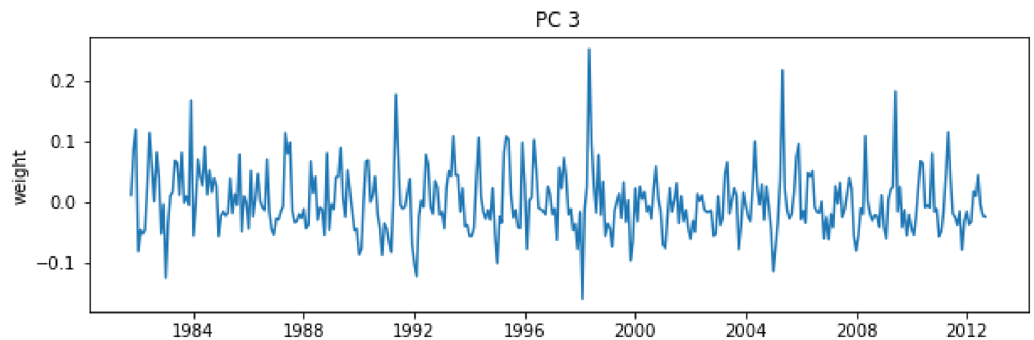


Spatial Weights (EOFs)

Temporal Weights (PCs)

Cold air pools

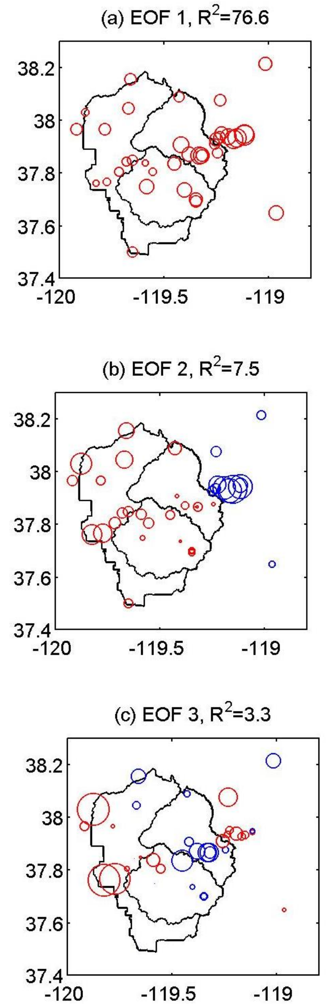




Reconstructing data from the top EOFs/PCs

$$X` = U`S`V`^T$$

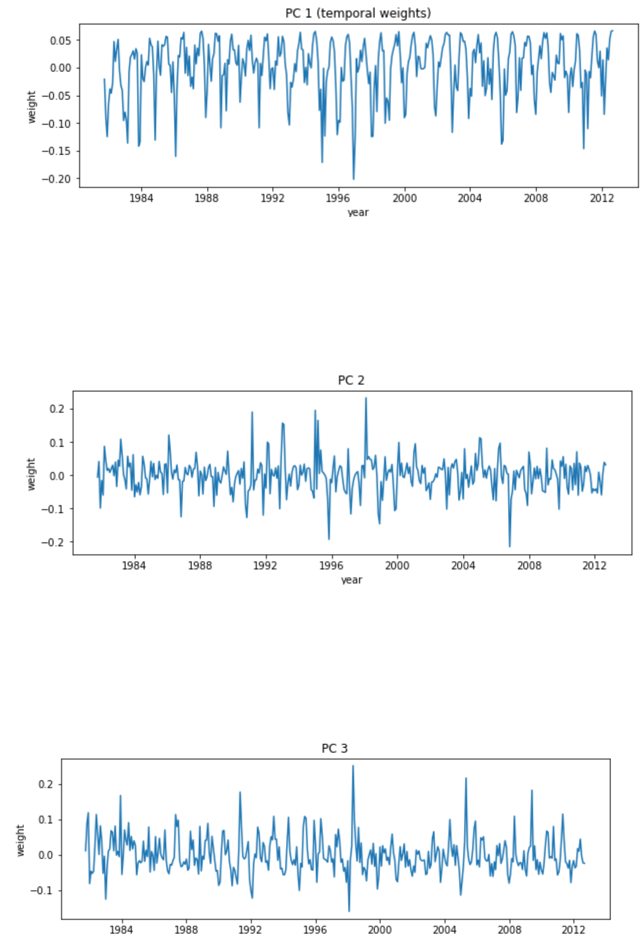
$X` =$



$\times$

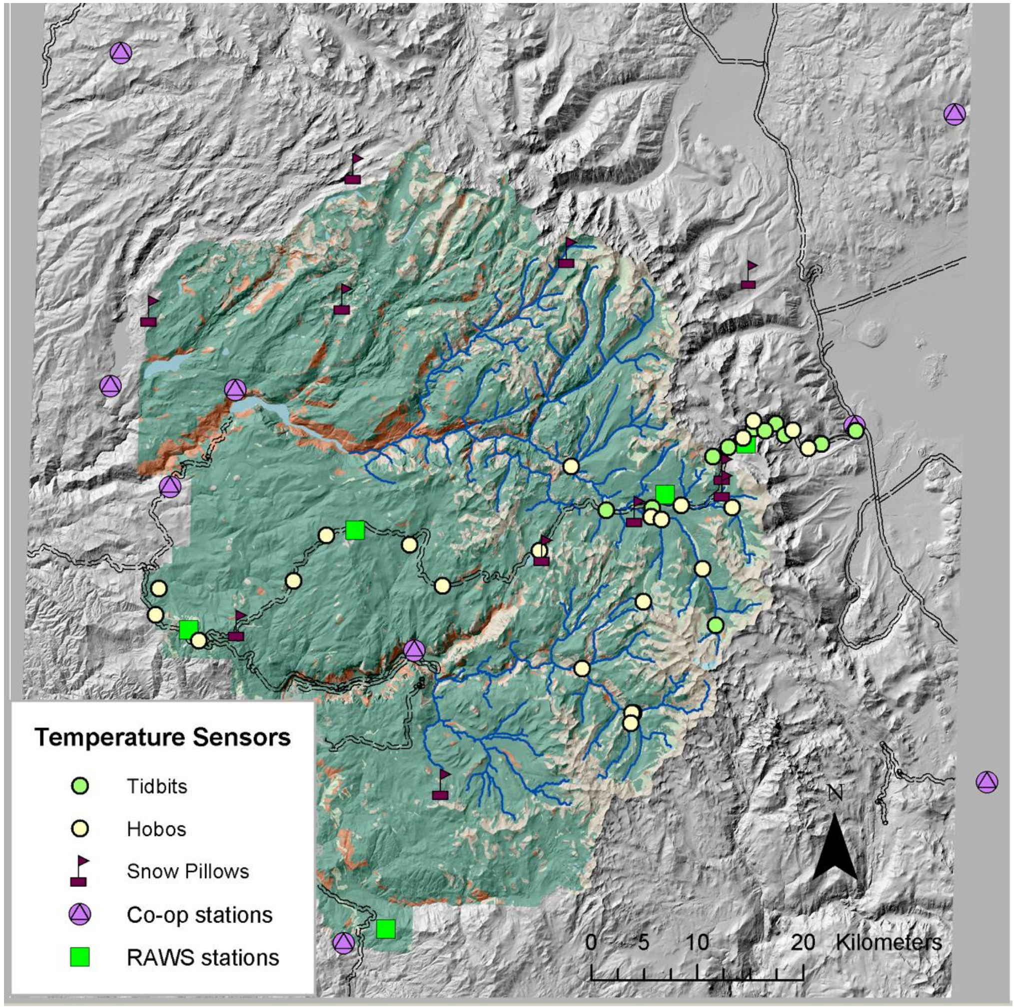
$$\begin{bmatrix} S_1 \\ S_2 \\ S_3 \end{bmatrix}$$

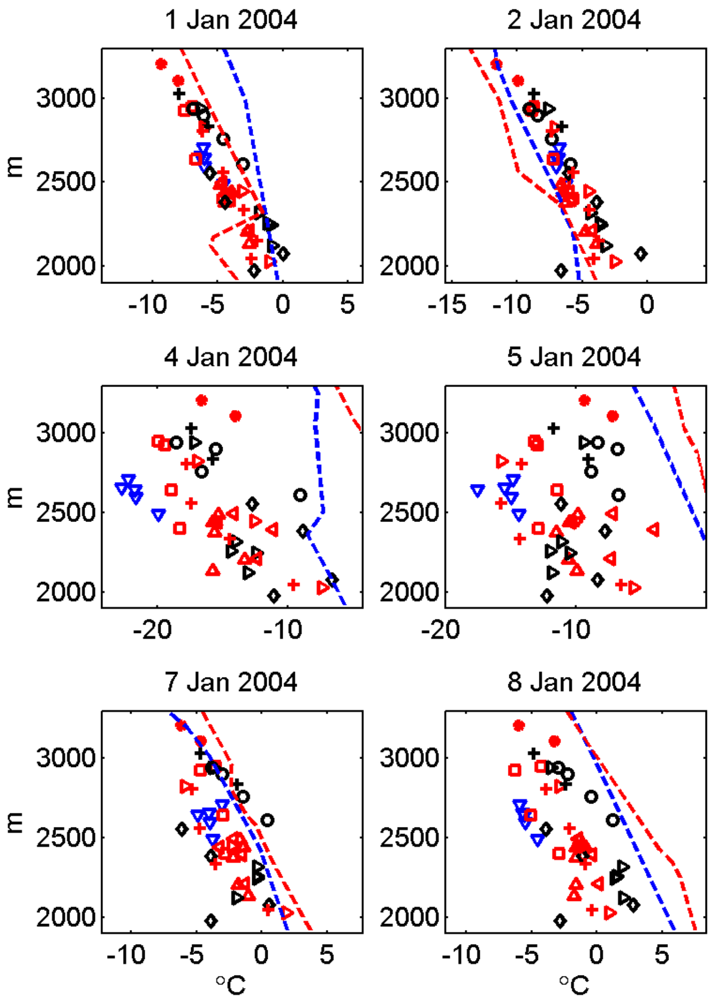
$\times$



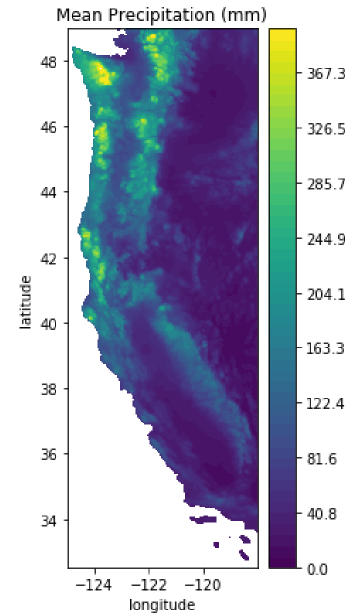
Use this information to map space and time separately:

- Use topography (DEM) to map spatial patterns (aspect, cold-air pools)
- Use readily available weather information to create timeseries of when and how much spatial patterns matter

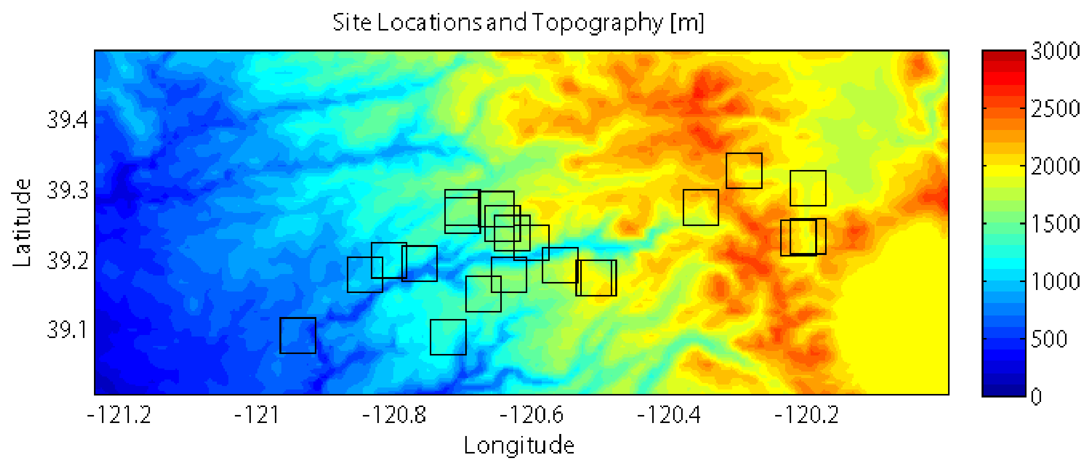




Lab 8-1: SVD with Monthly Precip.



Homework 7, problem 2 (see "Module 8"): Air temperature observations in complex terrain



https://mountain-hydrology-research-group.github.io/data-analysis/modules/module8.html